

3/5 answered

```

1 import java.util.*;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5
6         String[] product={"laptop","phone","headphone"};
7         double[] price={5000,2000,2000};
8
9         try{
10             int index=sc.nextInt();
11             int quantity=sc.nextInt();
12
13             System.out.println("product: "+product[index]);
14             if(quantity <=0)
15                 throw new IllegalArgumentException();
16             System.out.println("Total Cost: " +(price[index]*quantity));
17         }
18         catch(ArrayIndexOutOfBoundsException e){
19             System.out.println("Invalid product index");
20         }catch(InputMismatchException e){
21             System.out.println("Invalid numeric input");
22         }
23         catch(IllegalArgumentException e){
24             System.out.println("Invalid quantity");
25         }
26         System.out.println("Transaction completed");
27     }
28 }

```

1 2

Output

```

product: phone
Total Cost: 4000.0
Transaction completed

```

10 marks

3/5 answered

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             double balance=sc.nextDouble();
7             double withdrawal=sc.nextDouble();
8
9             if(withdrawal<=0)
10                 throw new IllegalArgumentException("Invalid withdraw");
11             if(withdrawal>balance)
12                 throw new InsufficientBalanceException("Insufficient");
13             balance-=withdrawal;
14
15             System.out.println("withdrawal succesfull");
16             System.out.println("Reamining Balance:"+ balance);
17
18         }
19         catch(InputMismatchException e){
20             System.out.println("Invalid numeric input");
21         }
22         catch(InsufficientBalanceException e){
23             System.out.println(e.getMessage());
24         }
25         catch(IllegalArgumentException e){
26             System.out.println(e.getMessage());
27         }
28         finally{
```

abc 3000

Output

```
Invalid numeric input
Transaction completed
```

Thread Priority
10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

3/5 answered

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String name=sc.next();
7             int age=sc.nextInt();
8             double temp=sc.nextDouble();
9             if(age<18)
10                 throw new InvalidAgeException("InvalidAgeException")
11             if(temp < 95 || temp>105)
12                 throw new InvalidTemperatureException("InvalidTemperatureException")
13             System.out.println("patient registered sucessfully");
14         }
15         catch(InvalidAgeException e){
16             System.out.println(e.getMessage());
17         }catch(InvalidTemperatureException e){
18             System.out.println(e.getMessage());
19         }
20         finally{
21             System.out.println("Registration completed");
22         }
23     }
24 }
25 class InvalidAgeException extends Exception{
26     InvalidAgeException (String msg){
27         super(msg);
28     }
29 }
```

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Output

patient registered sucessfully
Registration completed

Q5

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Your Code

```
1 class main{
2     public static void main(String[] args) throws InterruptedException
3
4         Emergency e=new Emergency();
5         General g=new General();
6         Billing b=new Billing();
7
8         e.setPriority(10);
9         g.setPriority(5);
10        b.setPriority(1);
11
12        e.start();
13        e.join();
14
15        g.start();
16        e.join();
17
18        b.start();
19        b.join();
20    }
21 }
22 class Emergency extends Thread{
23     public void run(){
24         System.out.println("Emergency patient processing");
25     }
26 }
27 class General extends Thread {
28     public void run(){
```

Input

stdin input

Output

```
Emergency patient processing
Billing processing
General patient processing
```

4/5 answered

5/5 answered

```
1 import java.util.*;
2 class main{
3     public static void main(String[] args) throws InterruptedException{
4         Order order = new Order();
5         Payment payment = new Payment();
6         Delivery delivery=new Delivery();
7
8         order.start();
9         order.join();
10
11        payment.start();
12        payment.join();
13
14        delivery.start();
15        delivery.join();
16    }
17 }
18 class Order extends Thread{
19     public void run(){
20         System.out.println("order processing");
21     }
22 }
23 class Payment extends Thread{
24     public void run(){
25         System.out.println("payment processing");
26     }
27 }
28 class Delivery extends Thread {
29     public void run(){
```

stdin input

Output

```
order processing
payment processing
Delivery Assignment
```